Graphs: Terminologies, Special Graphs, Representations

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Outline

1. Graph Introduction
2. Graph Terminology
3. Special Types of Graphs
4. Representing Graphs
5. Summary

Definition of a Graph

Definition 1: Graph

A graph $G = (V, E)$ consists of:

- V: a nonempty set of vertices (nodes)
- E: a set of edges

Each edge has either one or two vertices associated with it, called its **endpoints**.

Interpretation

If an edge is associated with $\{u, v\}$, we say that u and v are connected.

Graph Example: Network Model

Application Example

Graphs are widely used to model networks such as:

- Computer communication networks
- Transportation systems
- Social networks

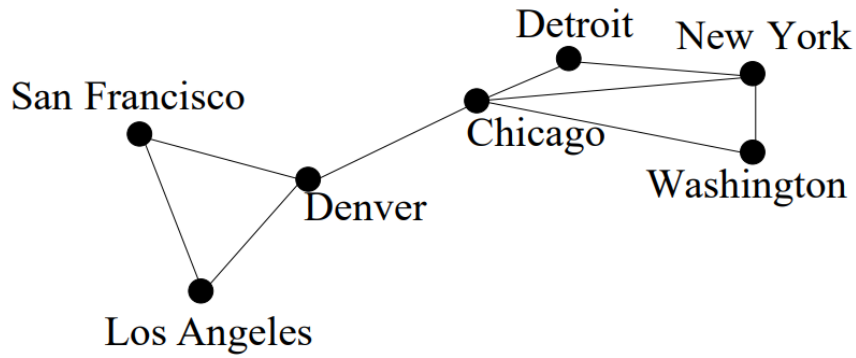


Figure: A computer network represented by a graph.

Loops and Multiple Edges

Special Edge Types

- An edge connecting a vertex to itself is called a **loop**
- Multiple edges may connect the same pair of vertices
- Such graphs are called **multigraphs**

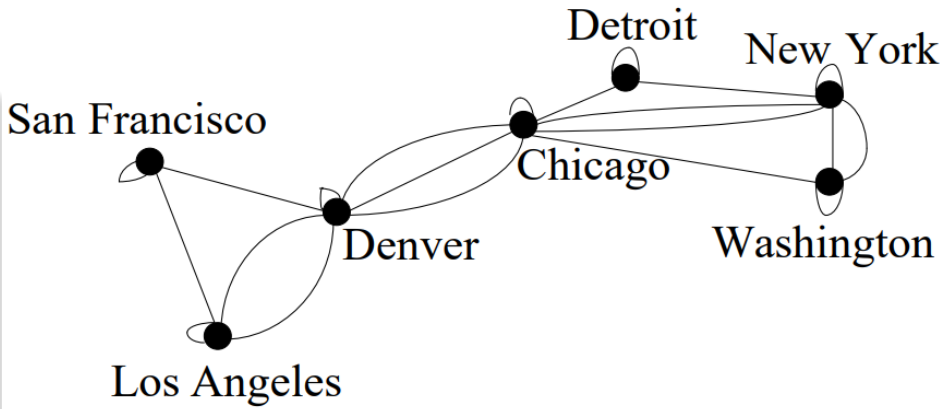


Figure: A graph representing a computer network with multiple edges and loops.

Directed Graphs

Definition 2: Directed Graph

A directed graph (digraph) $G = (V, E)$ consists of:

- A nonempty set V
- A set of directed edges (arcs)

Each directed edge is an ordered pair (u, v).

Terminology

- u = initial vertex
- v = terminal vertex

Directed Graph Example

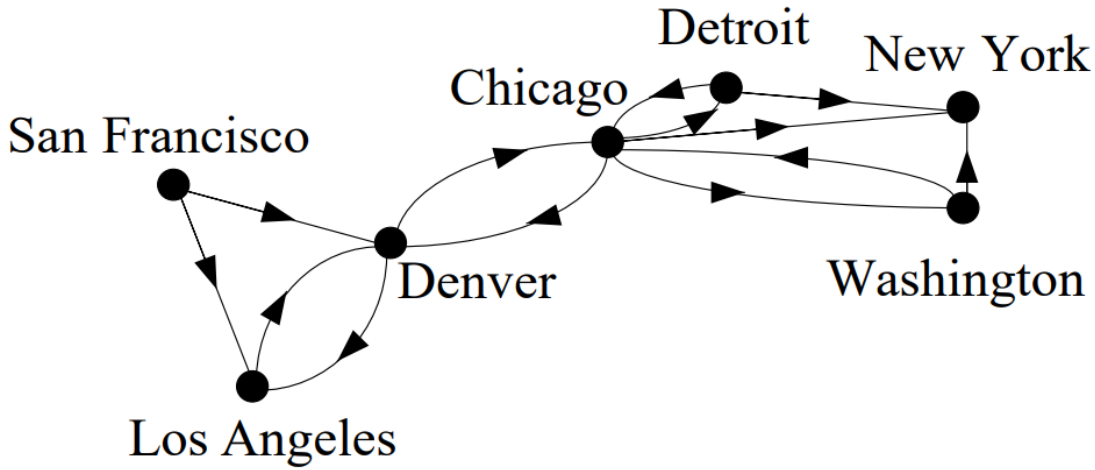


Figure: Communication network with one-way arrows.

Adjacency and Degree

Definition: Adjacent Vertices

Vertices u and v are adjacent if they are endpoints of the same edge.

Definition: Degree

The degree of vertex v in an undirected graph is:

$$\deg(v) = \text{number of incident edges}$$

A loop contributes 2 to the degree.

Handshaking Theorem

Theorem 1: Handshaking Theorem

Let $G = (V, E)$ be an undirected graph.

$$2|E| = \sum_{v \in V} \deg(v)$$

Example

If 10 vertices each have degree 6:

$$\sum \deg(v) = 60$$

$$2E = 60 \Rightarrow E = 30$$

In-Degree and Out-Degree

Definition

For directed graphs:

$$\deg^-(v) = \text{in-degree}$$

$$\deg^+(v) = \text{out-degree}$$

A loop contributes 1 to each.

Theorem

$$\sum \deg^-(v) = \sum \deg^+(v) = |E|$$

Complete Graphs

Definition

The complete graph K_n contains exactly one edge between each pair of distinct vertices.

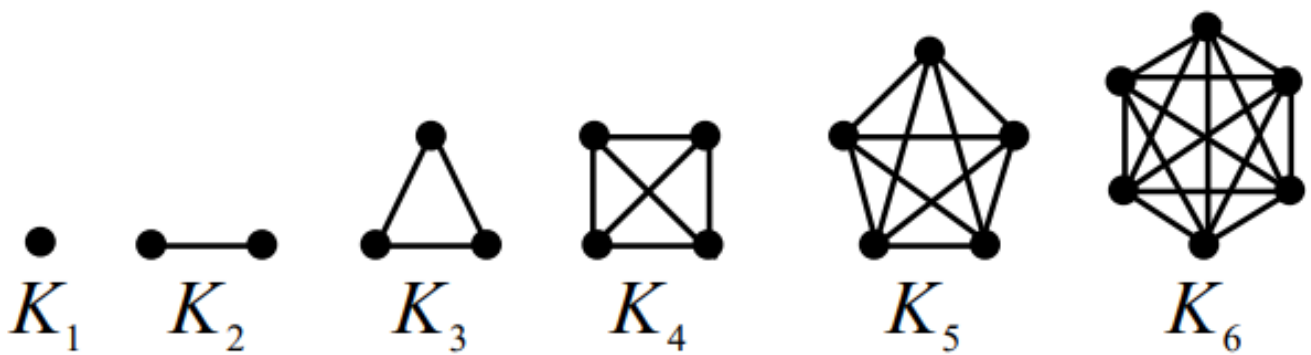


Figure: $K_1, K_2, K_3, K_4, K_5, K_6$ are all complete graphs.

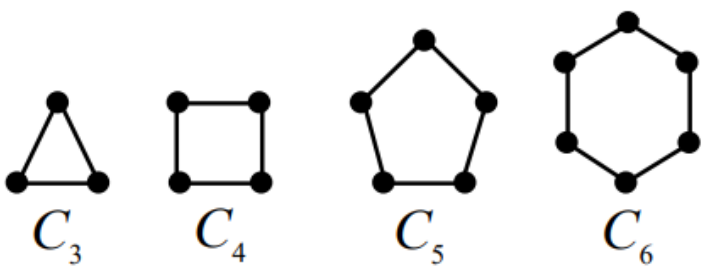
Cycles and Wheels

Cycle

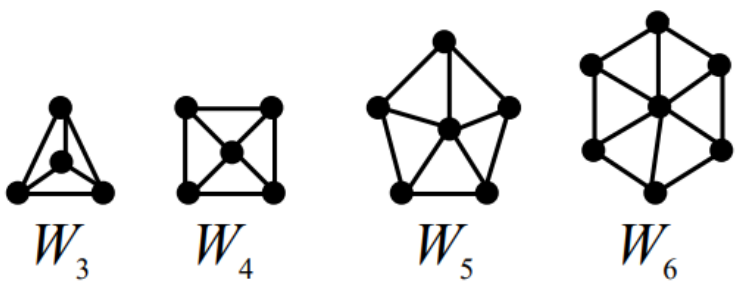
C_n consists of vertices forming a closed chain.

Wheel

W_n is formed by adding a central vertex to C_n.



Cycles (C_3, C_4, C_5, C_6)



Wheels (W_3, W_4, W_5)

Bipartite Graphs

Definition

A graph is bipartite if its vertices can be partitioned into V_1 and V_2 such that every edge connects a vertex in V_1 to one in V_2.

Examples

- C_6 is bipartite
- K_3 is not bipartite

Bipartite Example Figures

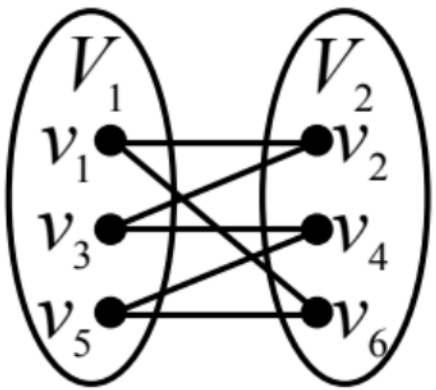


Figure: C_6 is bipartite as shown in Figure.



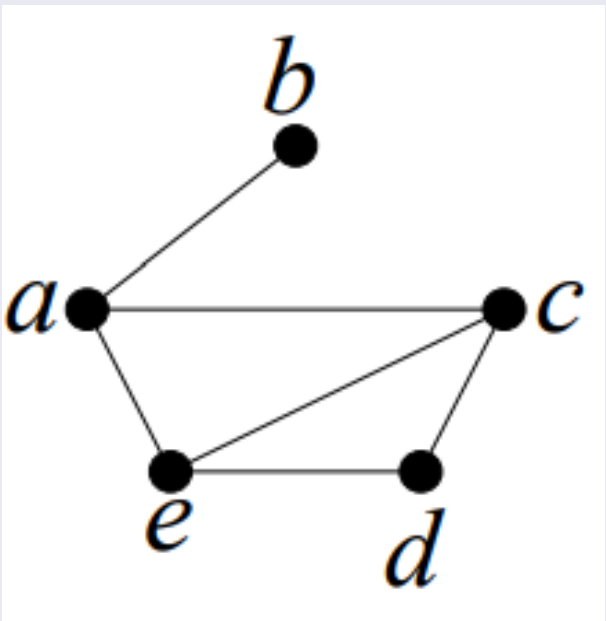
Figure: K_3 is not bipartite.

Adjacency Lists

Definition

An adjacency list represents each vertex together with the list of vertices adjacent to it.

Example



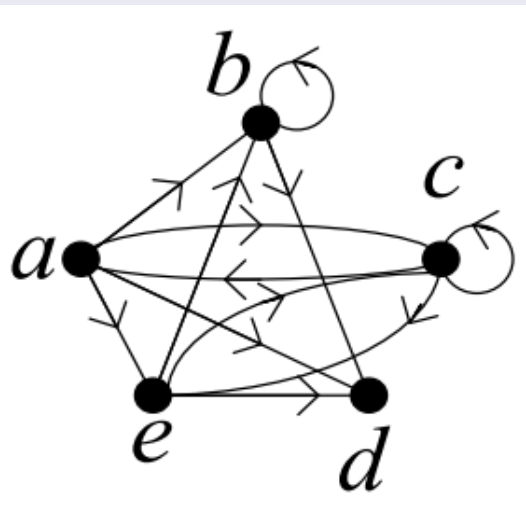
Vertex	Adjacent Vertices
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

Directed Graph Example

Problem

Represent the directed graph by listing all vertices that are the terminal vertices of edges starting at each vertex.

Edge List Representation



Initial Vertex	Terminal Vertices
a	b, c, d, e
b	b, d
c	a, c, e
d	
e	b, c, d

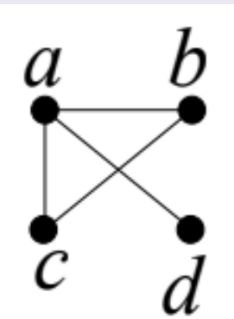
Adjacency Matrix (Undirected)

Definition

For $|V| = n$, the adjacency matrix is an $n \times n$ matrix:

$$a_{ij} = \begin{cases} 1 & \text{if } v_i \text{ adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

Example



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

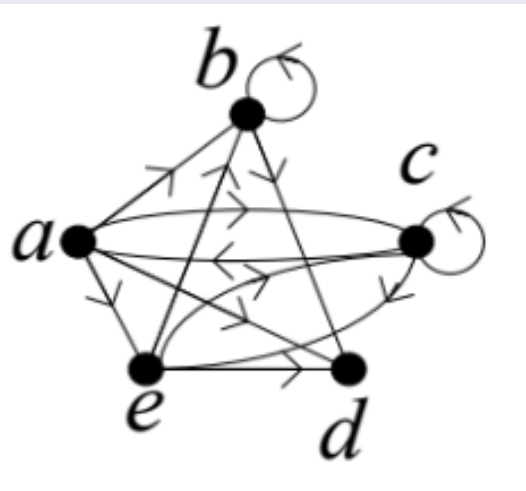
Adjacency Matrix (Directed)

Definition

For directed graphs:

$$a_{ij} = 1 \text{ if edge } (v_i, v_j) \text{ exists}$$

Example



$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Summary

- Graphs model relationships between objects
- Undirected and directed graphs capture different systems
- Degree concepts quantify connectivity
- Special graphs include complete, cycles, wheels, bipartite
- Graphs can be represented using adjacency lists and matrices

End of Lecture

Thank You